MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 8–10 (finishing P3)

Math Inquirers P3 Ages 8-10. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE builds fluency with the main idea of the lesson. Use it for warm-ups and for children who need confidence.
- THINK HARDER goes a step further: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are non-routine. They ask children to find every way, work backwards, test a case, justify an answer or notice what is missing. This is the style of thinking used in gifted-education-type problems. It is not an official GEP standard, and this program is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Number Sleuths (Numbers to 10 000)	6 + 8 + 9 questions
Lesson 2	Cryptarithm Crackers (Addition and Subtraction)	6 + 8 + 9 questions
Lesson 3	Times Table Treasure (Tables of 6, 7, 8 and 9)	6 + 8 + 9 questions
Lesson 4	Model Makers (Word Problems: Bar Models)	6 + 9 + 9 questions
Lesson 5	Remainder Riddles (Division)	6 + 8 + 9 questions
Lesson 6	Fraction Walls (Fractions)	6 + 8 + 9 questions
Lesson 7	Area Architects (Area and Perimeter)	6 + 8 + 9 questions
Lesson 8	Angle and Line Detectives (Angles, Perpendicular and Parallel Lines)	6 + 8 + 9 questions
Lesson 9	Measure and Money Masters (Length, mass, volume, money and time)	6 + 9 + 9 questions
Lesson 10	Challenge Day (Mixed puzzles: bar graphs and logic)	6 + 8 + 9 questions

Lesson 1: Number Sleuths

Numbers to 10 000 • Week 1

CORE

C1. Write the number: 6 thousands, 4 tens and 9 ones.

C2. What is the value of the digit 8 in 2837?

C3. Use the cards 9, 1, 4, 6 once each. Make the smallest 4-digit number.

C4. What number is 10 less than 5003?

C5. Continue the pattern: 7250, 7500, 7750, ___, ___

C6. Order from smallest to greatest: 4089, 4809, 4098, 4890

THINK HARDER

T1. Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.

T2. How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.

T3. A number has 7 thousands, 13 hundreds and 6 ones. What is the number?

T4. I am between 5000 and 6000. My hundreds, tens and ones digits are all the same. My digits add up to 11. What number am I?

T5. What is the smallest 4-digit number with all different digits that add up to 10?

T6. How many times do you write the digit 5 when you write all the numbers from 1 to 100?

T7. Continue the pattern: 1, 3, 7, 15, 31, ___, ___

T8. How many 4-digit numbers between 3000 and 4000 read the same forwards and backwards, like 3443?

CHALLENGE (non-routine: justify, find every way)

X1. Ali writes a 4-digit number using only odd digits, all different. It is as close to 5000 as possible. What is it?

X2. A pattern starts at 1000 and goes up by 7 each time: 1000, 1007, 1014, … Is 1100 in the pattern? Explain.

X3. How many digits are needed to number the pages of a book from page 1 to page 120?

X4. What is the smallest 4-digit number with all different digits that add up to 20?

X5. I am an even 4-digit number with all different digits. My thousands digit is 4 more than my ones digit. My hundreds digit is double my tens digit. My digits add up to 20. What number am I?

X6. Mei makes 3-digit numbers using any three of the cards 2, 4, 6, 8. How many different numbers can she make?

X7. How many 4-digit numbers have digits that add up to 4? Make a list in order.

X8. A 4-digit number has all different digits. Its thousands digit is the sum of its other three digits. What is the greatest such number? What is the smallest?

X9. Mei uses the cards 1, 2, 3, 4 once each to make 4-digit numbers. She lists them all from smallest to greatest. What is the 10th number in her list?

Lesson 2: Cryptarithm Crackers

Addition and Subtraction • Week 1

CORE

C1. Work out 3999 + 2456. Use a shortcut.

C2. 7000 − 2468 = ?

C3. Find the missing digit: 4□27 + 1365 = 5992

C4. Find the missing digit: 8□50 − 3420 = 5230

C5. In a magic square, every row adds up to 2400. One row has 900 and 600 in it. What is the missing number in that row?

C6. A shop had 5000 stickers. It sold 1275 on Monday and 1480 on Tuesday. How many stickers are left?

THINK HARDER

T1. Find the missing digits: 2□7□ + □6□9 = 6000

T2. In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200, □, □. The middle row is 2650, □. The top brick is 6000. Find the missing bricks.

T3. Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?

T4. What is the smallest 4-digit number whose digits add up to 30? What is the greatest?

T5. ◆● + ● = ●◆. Different shapes are different digits. Find ◆ and ●.

T6. Use the 9999 trick: 10 000 − 3746 = ?

T7. ▲▲ + ■■ = ●■●. Different shapes are different digits. Find them.

T8. Use the cards 2, 3, 5, 6 once each to make two 2-digit numbers. Make the difference as small as possible.

CHALLENGE (non-routine: justify, find every way)

X1. Find the missing digits: 8□0□ − □4□7 = 2468

X2. A 4-digit number has all different digits that add up to 30. Find the difference between the greatest and the smallest such number.

X3. Mei thinks of a number. She adds 2500, then takes away 1750, and gets 6000. What was her number?

X4. ■▲● − ●▲ = ●■▲. Different shapes are different digits. Find the digits.

X5. Make a magic square with the numbers 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700 and 1800. What is the magic total?

X6. Use the digits 1 to 9 once each: □□□ + □□□ = □□□. What is the greatest possible answer?

X7. ◆◆ + ●● + ▲▲ = ◆●▲. Different shapes are different digits. Find them.

X8. Use the digit cards 1 to 8 once each to make two 4-digit numbers. Make their difference as small as possible.

X9. ★■● + ★●■ = ●■★. Different shapes are different digits. Find them.

Lesson 3: Times Table Treasure

Tables of 6, 7, 8 and 9 • Week 2

CORE

C1. Find 8 × 6 and 6 × 8. What do you notice?

C2. You know 5 × 9 = 45. Use it to find 6 × 9.

C3. Fill in the box: 7 × □ = 63

C4. A spider has 8 legs. How many legs do 7 spiders have?

C5. Write the last digits of the 6 times table from 6 × 1 to 6 × 10. Which digits appear?

C6. Find 9 × 7 by working out 10 × 7 and taking away one 7.

THINK HARDER

T1. The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?

T2. What is the last digit of 8 × 8 × 8? Use last digits only.

T3. Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.

T4. Find ALL the 2-digit numbers that can be divided exactly by 7 and whose digits add up to 10.

T5. Which is more, 7 × 8 or 6 × 9? By how much?

T6. The last digit of 9 × 9 is 1. What is the last digit of 9 × 9 × 9 × 9?

T7. Ben multiplies a number by 6. His answer ends in 4. What could the last digit of his number be? Find all the possible digits.

T8. Mei's age × 6 = 48. Ali's age × 7 = 63. Who is older, and by how many years?

CHALLENGE (non-routine: justify, find every way)

X1. Four numbers in a multiplication square are 42 and 48 in the top row, and 49 and 56 in the row below. What are the two row numbers and the two column numbers?

X2. Ben writes the answers of the 6 times table from 6 × 1 to 6 × 20. How many of the answers end in 8?

X3. How many numbers from 1 to 200 can be divided exactly by both 6 and 9?

X4. Siti writes the 7 times table from 7 × 1 to 7 × 20. How many answers end in 3?

X5. Siti says the answer to 6 × 7 × 8 × 9 ends in 4. Is she right? Use last digits only.

X6. Mei multiplies two DIFFERENT numbers from 6, 7, 8 and 9. Find all the answers she can get.

X7. Mei has between 40 and 60 sweets. She can share them equally among 6 friends with none left over. She can also share them equally among 9 friends with none left over. How many sweets does she have?

X8. Ben keeps multiplying by 7: 7, 49, 343 ... He writes only the last digit each time: 7, 9, 3 ... What is the last digit when he has multiplied 8 sevens together? Explain the pattern.

X9. I am a 3-digit number less than 200. I can be divided exactly by 7 and by 9. My digits add up to 9. What number am I?

Lesson 4: Model Makers

Word Problems: Bar Models • Week 2

CORE

C1. There are 2345 people at a concert. 1180 are adults. How many are children? Draw a part-whole model.

C2. A red ribbon is 145 cm long. A blue ribbon is 68 cm longer. How long is the blue ribbon?

C3. Ravi has 512 stamps. Siti has 139 fewer stamps than Ravi. How many stamps do they have altogether?

C4. Ben has 4 times as many cards as Ali. Ali has 23 cards. How many cards does Ben have? How many do they have altogether?

C5. A box has 1000 beads. Mei uses 365 beads for a necklace and 248 beads for a bracelet. How many beads are left?

C6. A pen costs $3. A bag costs 5 times as much as the pen. How much more does the bag cost than the pen?

THINK HARDER

T1. Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?

T2. Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?

T3. Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?

T4. A bag of rice and a bag of flour weigh 9 kg altogether. The rice is 3 kg heavier than the flour. How heavy is each bag?

T5. Mei is 8 years old. Her mother is 36. How old will Mei be when her mother is 3 times as old as Mei?

T6. Ali has $45. Ben has $18. Grandma gives each boy the same amount of money. Now Ali has twice as much money as Ben. How much money did Grandma give each boy?

T7. Ben had 3 times as many marbles as Ravi. Ben gave Ravi 20 marbles, and then they had the same number. How many marbles did Ben have at first?

T8. A bag of sugar and 2 bags of flour have a total mass of 1700 g. Each bag of flour is 400 g heavier than the bag of sugar. What is the mass of the bag of sugar?

T9. Ali has saved $48. Ben has saved $15. From now on each boy saves $6 a week. After how many weeks will Ali have twice as much as Ben?

CHALLENGE (non-routine: justify, find every way)

X1. Ali and Ben have 120 stickers altogether. Ali has 3 times as many as Ben. Ali gives 20 stickers to Ben. How many stickers does each boy have now?

X2. A pencil box and 3 pens cost $25 altogether. The pencil box costs $5 more than one pen. How much does one pen cost?

X3. Ali, Ben and Ravi have 185 marbles altogether. Ali has twice as many as Ben. Ben has 15 more than Ravi. How many marbles does each boy have?

X4. Siti has 4 times as many beads as Mei. If Siti gives Mei 30 beads, they will have the same number. How many beads do they have altogether?

X5. A club has 3 times as many boys as girls. Then 12 boys leave and 12 girls join, and now there are the same number of boys and girls. How many children are in the club?

X6. 2 jugs and 3 cups hold 1800 ml of water altogether. A jug holds 3 times as much as a cup. How much does a cup hold?

X7. Ravi is 7 years old and his sister is 3. In how many years will their ages add up to 30?

X8. A pen costs $3 more than a ruler. Mei buys 2 pens and 2 rulers for $18. How much does one pen cost?

X9. A ribbon 4 m 75 cm long is cut into 3 pieces. The second piece is twice as long as the first. The third piece is 25 cm longer than the second. How long is the third piece?

Lesson 5: Remainder Riddles

Division • Week 3

CORE

C1. Work out 35 ÷ 4. Write the answer and the remainder.

C2. Work out 58 ÷ 7. Write the answer and the remainder.

C3. 6 friends share 45 stickers equally. How many does each friend get? How many are left over?

C4. 30 children go on an outing. Each car holds 4 children. How many cars are needed?

C5. A ribbon is 50 cm long. Siti cuts it into pieces 6 cm long. How many whole pieces does she get? How much ribbon is left?

C6. Ben packs muffins into boxes of 5. He fills 7 boxes and has 3 muffins left over. How many muffins does he have?

THINK HARDER

T1. Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?

T2. I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?

T3. What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?

T4. A boat can carry 9 people at a time. 100 people want to cross the river. How many boatloads are needed?

T5. Siti fills 9 bags with 7 oranges each and has 5 oranges left over. She packs all the oranges again into bags of 8. How many full bags does she get? How many are left over?

T6. A pattern goes circle, triangle, square, circle, triangle, square, … What is the 50th shape?

T7. Find all the 2-digit numbers less than 40 that leave a remainder of 2 when divided by 3 AND when divided by 4.

T8. A pen costs $4. Ravi has $35. How many pens can he buy? How much more money does he need to buy one more pen?

CHALLENGE (non-routine: justify, find every way)

X1. I am a 2-digit number. When I am divided by 3, by 4 or by 5, the remainder is always 2. What number am I?

X2. A box has more than 30 but fewer than 50 chocolates. They can be shared equally among 2, 3, 4 or 6 children with none left over. Find all the numbers of chocolates there could be.

X3. The letters M A T H M A T H M A T H … go on and on. What is the 75th letter?

X4. Ben counts his marbles in 5s and has 3 left over. He counts them in 7s and has 3 left over. He has more than 50 and fewer than 100 marbles. How many does he have?

X5. When 100 is divided by a 1-digit number, the remainder is 1. Find all the numbers it could be.

X6. A farmer counts her eggs. In 2s, 3s, 4s, 5s or 6s there is always 1 egg left over. In 7s there are none left over. What is the smallest number of eggs she can have?

X7. A number is divided by 6. The answer and the remainder are the same number, and it is not 0. Find all the numbers it could be.

X8. Today is Monday. What day of the week will it be 100 days from today?

X9. Mr Lim gives pens to his pupils. If each pupil gets 4 pens, 10 pens are left over. If each pupil gets 6 pens, he is 4 pens short. How many pupils are there? How many pens does he have?

Lesson 6: Fraction Walls

Fractions • Week 3

CORE

C1. Fill in the box: 1/2 = ?/10

C2. Which is bigger: 1/6 or 1/9?

C3. Work out 2/9 + 5/9.

C4. Work out 1 − 3/8.

C5. Put these in order, smallest first: 5/12, 1/12, 11/12.

C6. Fill in the box: 2/5 = ?/10

THINK HARDER

T1. Which is bigger, 3/5 or 7/10? Show how you know.

T2. Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?

T3. Put these in order, smallest first: 2/3, 5/6, 7/12.

T4. Write two fractions equal to 4/6: one with a smaller bottom number and one with a bigger bottom number.

T5. A jug is 5/6 full of water. Ravi pours out 1/2 of the whole jug. What fraction of the jug is still full?

T6. Which fraction with bottom number 8 is exactly halfway between 1/4 and 1/2?

T7. Work out 1/2 + 1/4 + 1/8.

T8. Mei says 2/3 is bigger than 3/4, because 2/3 is missing only one piece. Is she right? Explain.

CHALLENGE (non-routine: justify, find every way)

X1. Which is bigger, 3/7 or 5/9? Use 1/2 to help you.

X2. Which is bigger, 2/5 or 4/9? Hint: make the top numbers the same.

X3. Use the digits 1, 2, 3 and 6, once each, to make two equal fractions that are each less than 1: ?/? = ?/?. Find both ways.

X4. A ribbon is cut into equal pieces. Ali takes 1/3 of the ribbon and Ben takes 1/6 of it. Mei takes the rest. Who has the most ribbon?

X5. Find all the fractions with bottom number 12 that are bigger than 2/3 but smaller than 1.

X6. I am a fraction equal to 3/4. My top number and my bottom number add up to 14. What fraction am I?

X7. A fraction is equal to 2/3. Its bottom number is 4 more than its top number. What is the fraction?

X8. Fill one whole bar using exactly 5 pieces. You may use 1/2, 1/4 and 1/8 pieces, as many of each as you like. Find all the ways.

X9. Siti and Ravi each have the same size bottle of juice. Siti drinks 2/3 of her bottle. Ravi drinks 7/12 of his. Who drinks more? How much more?

Lesson 7: Area Architects

Area and Perimeter • Week 4

CORE

C1. Find the perimeter of a rectangle 9 cm long and 4 cm wide.

C2. Find the area of a rectangle 7 cm long and 6 cm wide.

C3. A square has a perimeter of 32 cm. How long is each side?

C4. A square has an area of 36 cm². What is its perimeter?

C5. A rectangle has an area of 40 cm². Its length is 8 cm. Find its breadth and its perimeter.

C6. A figure is made of 7 squares, each 1 cm by 1 cm. What is its area? Can you say its perimeter without seeing it?

THINK HARDER

T1. Find all the rectangles with whole-cm sides and an area of 24 cm². Which one has the shortest perimeter?

T2. A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.

T3. A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?

T4. A square with sides of 6 cm is cut into two rectangles of the same size. What is the perimeter of each rectangle?

T5. How many tiles 2 cm by 2 cm are needed to cover a mat 10 cm by 8 cm?

T6. A rectangle has an area of 18 cm² and a perimeter of 18 cm. Find its length and breadth.

T7. Mei's garden is 9 m by 5 m. Ali's garden is 7 m by 7 m. Whose garden needs the longer fence? Whose has the bigger area?

T8. Use exactly 6 squares (each 1 cm by 1 cm), whole sides touching. Draw a figure with a perimeter of 10 cm and one with a perimeter of 14 cm. What is the biggest perimeter possible?

CHALLENGE (non-routine: justify, find every way)

X1. Find all the rectangles with whole-cm sides and a perimeter of 24 cm. Which has the biggest area?

X2. A rectangle is made of 2 same-size squares side by side. Its perimeter is 36 cm. What is its area?

X3. A rectangle 6 cm by 4 cm is cut into two rectangles with one straight cut that is parallel to one side. Find the total perimeter of the two pieces. Is there more than one answer?

X4. A square has a perimeter of 20 cm. Four of these squares are put together to make one big square. What is the perimeter of the big square?

X5. Each small square is 1 cm by 1 cm. Draw a figure of 7 squares with a perimeter of 12 cm, and another with a perimeter of 16 cm.

X6. The area of a rectangle is 36 cm². Its length is 4 times its breadth. Find its perimeter.

X7. A rectangle is made of 3 same-size squares in a row. Its perimeter is 32 cm. What is its area?

X8. Ali joins 5 same-size squares with sides of 2 cm, whole sides touching. What is the biggest perimeter his shape can have? The smallest?

X9. A rectangle 7 cm by 5 cm has a 1 cm square cut out of each of its 4 corners. Find the new area and the new perimeter.

Lesson 8: Angle and Line Detectives

Angles, Perpendicular and Parallel Lines • Week 4

CORE

C1. How many right angles does a rectangle have?

C2. At 3 o'clock, what angle do the hands of a clock make?

C3. At 2 o'clock, is the angle between the clock hands smaller or bigger than a right angle?

C4. Name 2 capital letters that have a pair of parallel lines.

C5. Two straight lines cross like a plus sign (+). How many right angles do they make?

C6. 3 lines go out from one point, spread out like a fan. How many angles can you find?

THINK HARDER

T1. I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?

T2. A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?

T3. At which o'clock times do the hands of a clock make a right angle?

T4. The # sign is made of 4 lines: 2 upright lines and 2 lines across. How many pairs of parallel lines does it have? How many pairs of perpendicular lines?

T5. Draw me on square grid paper: (a) I have 4 sides, exactly 2 right angles and exactly 1 pair of parallel lines. (b) I have 4 sides and 2 pairs of parallel lines, but no right angles.

T6. How many right angles can you find in the letter E (4 straight lines)?

T7. Is the smaller angle between the clock hands at 4 o'clock the same size as at 8 o'clock?

T8. How many pairs of parallel lines does a rectangle have? How many pairs of perpendicular lines?

CHALLENGE (non-routine: justify, find every way)

X1. 4 lines from one point make 6 angles. How many lines from one point are needed to make exactly 10 angles?

X2. The minute hand turns from 12 to 3. How many right angles has it turned through? From 12 to 9 (going clockwise)?

X3. A square has 2 lines down and 2 lines across inside it, like a noughts-and-crosses grid. How many right angles can you find?

X4. How many right angles are there in the letter F (3 straight lines)?

X5. The letter E is made of 4 lines. List every pair of lines. How many pairs are there altogether?

X6. Ben says: "A shape with 4 sides and 2 pairs of parallel lines must have right angles." Is he right?

X7. A 4-sided shape has 3 right angles. Draw it. Test the 4th angle with your checker. Can a 5-sided shape have exactly 3 right angles? Draw one.

X8. A long rectangle is cut into 6 squares in a row by 5 lines. How many right angles can you find?

X9. Can a 6-sided shape have 6 right angles inside it? What is the most right angles a 6-sided shape can have inside?

Lesson 9: Measure and Money Masters

Length, mass, volume, money and time • Week 5

CORE

C1. 2 m 45 cm = ____ cm

C2. 3060 g = ____ kg ____ g

C3. A pail holds 4 ℓ 500 ml. A jug holds 1 ℓ 750 ml. How much more does the pail hold?

C4. Ali buys a sandwich for $3.80 and a drink for $1.45. He pays with a $10 note. How much change does he get?

C5. Write 6:20 p.m. in 24-hour time.

C6. A lesson starts at 10:45 a.m. and ends at 12:05 p.m. How long is it?

THINK HARDER

T1. A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?

T2. A shop opens at 09:30 and closes at 21:00. How long is it open?

T3. A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?

T4. Buses leave the interchange every 25 minutes. The first bus leaves at 06:40. The ride to school takes 35 minutes. Ravi must reach school by 08:30. Which is the latest bus he can take?

T5. Ben has only 20¢ and 50¢ coins. He has 9 coins worth $3 altogether. How many of each coin does he have?

T6. A night train leaves at 22:40 and arrives at 01:15 the next morning. How long is the trip?

T7. Siti's home is 1 km 250 m from school. She walks there and back every day. How far does she walk in 2 days?

T8. Ali needs 2 ℓ of milk. A 250 ml carton costs $1.10 and a 1 ℓ carton costs $3.60. What is the cheapest way to buy 2 ℓ?

T9. Mei goes to sleep at 21:45 and wakes up at 06:30. How long does she sleep?

CHALLENGE (non-routine: justify, find every way)

X1. A 5 m ribbon is cut into two pieces. One piece is 80 cm longer than the other. How long is each piece?

X2. You have a 300 ml cup, a 500 ml cup (no marks on either) and a tap. How can you get exactly 100 ml in a cup?

X3. Ravi has $10. He buys some erasers at 85¢ each and gets $1.50 change. How many erasers does he buy?

X4. A pail has 4 ℓ 200 ml of water. Ali fills 3 bottles that hold 900 ml each. He pours the rest equally into 2 jugs. How much water is in each jug?

X5. School starts at 08:15. There are 5 lessons of 35 minutes and a 20-minute recess after the 3rd lesson. When does the last lesson end?

X6. Ali and Ben have $24 altogether. After Ali spends $3.60, they have the same amount of money. How much did Ali have at first?

X7. A bottle full of water weighs 1 kg 200 g. After half of the water is poured out, it weighs 750 g. How heavy is the empty bottle?

X8. Mei buys 3 pens and 2 files for $9.20. Each file costs 60¢ more than each pen. How much does one pen cost?

X9. The clock at the station is 10 minutes fast. The train leaves when this clock shows 13:05. The trip takes 1 h 40 min. What is the real (correct) time when the train arrives? Give it in 24-hour and 12-hour time.

Lesson 10: Challenge Day

Mixed puzzles: bar graphs and logic • Week 5

CORE

C1. Mei is 5th from the front and 4th from the back of a queue. How many people are in the queue?

C2. Two numbers add up to 50. One number is 12 more than the other. What are the numbers?

C3. Make all the 3-digit numbers you can with the digit cards 1, 4 and 7 (use each card once). Which of them are even?

C4. A bar graph has lines at 0, 10, 20, 30, 40 and 50. A bar stops halfway between 30 and 40. What number does it show?

C5. 5, 9, 13, 17, … What is the 10th number in this pattern?

C6. Ali has $5. He buys 2 drinks at $1.35 each. How much money does he have left?

THINK HARDER

T1. In a queue, there are 4 people in front of Ravi and 7 people behind him. How many people are in the queue?

T2. Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?

T3. Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?

T4. Use the digits 3, 0, 6 and 9 once each. Make the largest even 4-digit number and the smallest odd 4-digit number.

T5. A bar graph shows stamps: Ali 35 and Ben 20. Mei has 15 more than Ben. Siti has as many as Ali and Ben together. How many stamps do the four children have altogether?

T6. Ali, Ben and Siti stand in a line. Ali is not first. Ben is just behind Siti. Who is first?

T7. A snail climbs a wall 12 m high. Each day it climbs up 5 m, and each night it slides down 2 m. On which day does it reach the top?

T8. Two numbers add up to 100. Their difference is 24. What are the numbers?

CHALLENGE (non-routine: justify, find every way)

X1. Mei, Ali and Ravi have 90 stickers altogether. Ali has 10 more than Mei. Ravi has 5 fewer than Ali. How many stickers does each child have?

X2. How many squares of any size are there in a 3 by 3 grid of squares?

X3. 6 children meet. Each child shakes hands once with every other child. How many handshakes are there?

X4. How many 2-digit numbers have digits that add up to 9?

X5. A bar graph shows the books 4 children read. Ali's and Ben's bars are the same height. Mei read 3 more than Ali. Siti read 2 fewer than Mei. Altogether they read 44 books. How many did each read?

X6. Siti has 8 coins. They are all 10¢ and 50¢ coins, worth $2 altogether. How many of each coin does she have?

X7. One of Ali, Ben and Mei ate the last kueh. Ali says, "Ben ate it." Ben says, "I did not eat it." Mei says, "I did not eat it." Only ONE of them is telling the truth. Who ate the kueh?

X8. In a queue, Siti is 6th from the front and Ravi is 4th from the back. There are 3 children between them, and Siti is in front of Ravi. How many children are in the queue?

X9. A bar graph shows the paper collected by 4 classes. 3P1 collected 10 kg more than 3P2. 3P3 collected the same as 3P1. 3P4 collected 5 kg less than 3P2. Altogether they collected 135 kg. How much did each class collect?

Answer key (tutor only)

Lesson 1: Number Sleuths

C1. 6049
C2. 800
C3. 1469
C4. 4993
C5. 8000, 8250 (add 250)
C6. 4089, 4098, 4809, 4890
T1. Greatest even: 8520. Smallest odd: 2085 (5 is the only odd card, so it goes last).
T2. 3 numbers: 1001, 1010, 1100.
T3. 8306 (7000 + 1300 + 6)
T4. 5222 (5 + 2 + 2 + 2 = 11)
T5. 1027 (start 1, 0; the last two digits must add to 9 and cannot be 0 or 1, so 2 and 7)
T6. 20 (ten times in the ones place: 5, 15, ..., 95; ten times in the tens place: 50 to 59)
T7. 63, 127 (double and add 1; the jumps are 2, 4, 8, 16, 32, 64)
T8. 10 (3003, 3113, 3223, ..., 3993)
X1. 5137 (137 more than 5000). The closest below 5000 is 3975, which is 1025 less.
X2. No. 14 jumps of 7 is 98, so 1098 is in the pattern. The next number is 1105, so 1100 is skipped.
X3. 252 digits: pages 1–9 use 9, pages 10–99 use 90 × 2 = 180, pages 100–120 use 21 × 3 = 63.
X4. 1289 (1 then 0 fails: the last two would need to add to 19; 1 then 2 leaves 17 = 8 + 9)
X5. 6842 (6 = 2 + 4, 8 = 2 × 4, 6 + 8 + 4 + 2 = 20). It is the only one.
X6. 24 (6 numbers start with each card: for 2, the numbers are 246, 248, 264, 268, 284, 286)
X7. 20. Starting with 4: 1; with 3: 3; with 2: 6; with 1: 10 (1, 3, 6, 10: the jumps go up by 1 each time).
X8. Greatest 9810 (9 = 8 + 1 + 0). Smallest 3012 (the smallest sum of three different digits is 0 + 1 + 2 = 3).
X9. 2341. Six numbers start with 1; then 2134 (7th), 2143 (8th), 2314 (9th), 2341 (10th).

Lesson 2: Cryptarithm Crackers

C1. 6455 (add 4000 to get 6456, then take away 1)
C2. 4532
C3. 6 (4627 + 1365 = 5992)
C4. 6 (8650 – 3420 = 5230)
C5. 900
C6. 2245
T1. 2371 + 3629 = 6000 (the only way)
T2. Bottom row 1200, 1450, 1900. Middle row 2650, 3350.
T3. 5643 (he added 900 too much: 6543 – 900)
T4. Smallest 3999, greatest 9993.
T5. ◆ = 8, ● = 9: 89 + 9 = 98
T6. 6254 (9999 – 3746 = 6253, then add 1)
T7. ▲ = 9, ■ = 2, ● = 1: 99 + 22 = 121
T8. 9 (35 – 26, or 62 – 53)
X1. 8905 – 6437 = 2468 (the only way; check 2468 + 6437 = 8905)
X2. 3087. The only four different digits that add up to 30 are 9, 8, 7, 6, so 9876 – 6789 = 3087.
X3. 5250 (work backwards: 6000 + 1750 = 7750, 7750 – 2500 = 5250)
X4. ■ = 7, ▲ = 3, ● = 6: 736 – 63 = 673
X5. 4200, with 1400 in the centre. One answer: 1100, 1600, 1500 / 1800, 1400, 1000 / 1300, 1200, 1700.
X6. 981, e.g. 235 + 746 = 981
X7. ◆ = 1, ● = 9, ▲ = 8: 11 + 99 + 88 = 198

X8. 247: 5123 – 4876 (the only way). The thousands digits must be next to each other; 5 and 4 in the middle work best.

X9. ★ = 4, ■ = 5, ● = 9: 459 + 495 = 954

Lesson 3: Times Table Treasure

C1. Both are 48. Swapping the numbers does not change the answer.

C2. 54 (one more 9: 45 + 9).

C3. 9

C4. 56

C5. 6, 2, 8, 4, 0, 6, 2, 8, 4, 0. Only 6, 2, 8, 4 and 0.

C6. 70 – 7 = 63

T1. 54 and 72 (5 + 4 = 9 and 7 + 2 = 9; 6 + 5 and 8 + 3 make 11). 54 = 9 × 6, 72 = 9 × 8.

T2. 2. 8 × 8 = 64 ends in 4, and 4 × 8 = 32 ends in 2. (8 × 8 × 8 = 512.)

T3. Yes. 6 groups can always be put into 3 pairs of groups, so nothing is left over. (6 × 7 = 3 × 14 = 42.)

T4. 28 and 91. (List 14, 21, 28 ... 98 and add the digits of each: only 28 and 91 make 10.)

T5. 7 × 8 = 56 is 2 more than 6 × 9 = 54.

T6. 1. Last digits go 9, 1, 9, 1. (9 × 9 × 9 × 9 = 6561.)

T7. 4 or 9 (6 × 4 = 24 and 6 × 9 = 54; no other digit gives a last digit of 4).

T8. Mei is 8, Ali is 9, so Ali is older by 1 year.

X1. Rows 6 and 7, columns 7 and 8. Start with 49: it can only be 7 × 7.

X2. 4 (18, 48, 78, 108). The last digits 6, 2, 8, 4, 0 repeat every 5, so 8 comes once in every 5.

X3. 11. They are the numbers in the 18 times table: 18, 36, 54 ... 198.

X4. 2 (63 and 133). Each digit appears once in every 10 answers.

X5. Yes. 6 × 7 = 42 → 2; 2 × 8 = 16 → 6; 6 × 9 = 54 → 4. (The answer is 3024.)

X6. 6 answers: 42, 48, 54, 56, 63, 72 (6×7, 6×8, 6×9, 7×8, 7×9, 8×9).

X7. 54. Numbers that work for 6 and 9 are 18, 36, 54, 72 ...; only 54 is between 40 and 60.

X8. 1. The last digits go 7, 9, 3, 1 and then repeat, so the 4th and the 8th are both 1.

X9. 126. Numbers that work for 7 and 9 are 63, 126, 189 ...; the 3-digit ones below 200 are 126 and 189, and only 1 + 2 + 6 = 9.

Lesson 4: Model Makers

C1. 1165

C2. 213 cm

C3. Siti 373; altogether 885

C4. Ben 92; altogether 115

C5. 387

C6. $12 (the bag costs $15; 4 more units of $3)

T1. 48. 4 units = 96, 1 unit = 24, and the difference is 2 units.

T2. 27 (half of the difference).

T3. 250. Take away the extras (50 + 100 = 150): 450 ÷ 3 = 150 in Box B, so A has 200 and C has 250.

T4. Flour 3 kg, rice 6 kg (9 – 3 = 6, 6 ÷ 2 = 3).

T5. 14. The difference, 28 years, never changes. Mother 3 units, Mei 1 unit, so 2 units = 28 and Mei is 14 (mother 42).

T6. $9. The difference, $27, stays the same. Now Ali has 2 units and Ben 1 unit, so 1 unit = $27. Ben now has $27, so Grandma gave each boy $27 – $18 = $9. Check: $54 = 2 × $27.

T7. 60. The difference is 2 units and Ben gave half of it, so 1 unit = 20. Ravi had 20, Ben had 60.

T8. 300 g (1700 – 800 = 900 for 3 equal units, 900 ÷ 3 = 300).

T9. 3 weeks. The difference, $33, stays the same, so Ben must reach $33: $15 + $18. Then Ali has $66.

X1. Ben 30, Ali 90 at first (4 units = 120); after giving: Ali 70, Ben 50.

X2. $5 (take away the extra $5: 4 equal units = $20). The box costs $10.

X3. Ravi 35, Ben 50, Ali 100.

X4. 100. The difference is 3 units = 60, so 1 unit = 20: Mei 20, Siti 80.

X5. 48 (12 girls and 36 boys at first; 24 and 24 after).

X6. 200 ml. 2 jugs = 6 cups, so 9 cups = 1800 ml.

X7. 10 years. Now they add up to 10, and each year the total grows by 2. (17 + 13 = 30.)

X8. $6. 2 pens and 2 rulers is 2 pairs, so one pen and one ruler cost $9. The pen is $3 more, so take away $3: $6 is 2 equal parts, a ruler costs $3 and a pen costs $6. Check: 2 × 6 + 2 × 3 = 18.

X9. 2 m 5 cm. 475 – 25 = 450 cm for 5 equal units, so 1 unit = 90 cm; third piece = 180 + 25 = 205 cm.

Lesson 5: Remainder Riddles

C1. 8 r 3 (4 × 8 = 32, 35 – 32 = 3)

C2. 8 r 2 (7 × 8 = 56)

C3. 7 each, 3 left over (6 × 7 = 42)

C4. 8 cars (30 ÷ 4 = 7 r 2; the 2 children left over need a car too)

C5. 8 pieces, 2 cm left (6 × 8 = 48)

C6. 38 (7 × 5 + 3)

T1. 6 bracelets (53 ÷ 8 = 6 r 5). She needs 3 more beads (56 – 53).

T2. 23 (remainder 3 when ÷ 4: 23 and 27; 23 ÷ 6 = 3 r 5, 27 ÷ 6 = 4 r 3)

T3. 98 (9 × 10 + 8 = 98; 99 ÷ 9 has no remainder)

T4. 12 boatloads (100 ÷ 9 = 11 r 1; the last person needs a boatload too)

T5. 68 oranges (9 × 7 + 5). 68 ÷ 8 = 8 r 4: 8 full bags, 4 left over.

T6. Triangle (50 ÷ 3 = 16 r 2; the 2nd shape in the repeat)

T7. 14, 26 and 38 (each is 2 more than a number in both the 3 and 4 tables: 12, 24, 36)

T8. 8 pens (35 ÷ 4 = 8 r 3). He needs $1 more.

X1. 62 (60 is in the 3, 4 and 5 tables; 60 + 2 = 62; the next one, 122, has 3 digits)

X2. 36 and 48 (numbers in the 12 times table)

X3. T (75 ÷ 4 = 18 r 3; the 3rd letter of MATH)

X4. 73 (35 is in both tables; 35 + 3 = 38 is too small; 70 + 3 = 73)

X5. 3 and 9 (100 ÷ 3 = 33 r 1, 100 ÷ 9 = 11 r 1; 99 is only in the 1, 3 and 9 tables, and dividing by 1 leaves no remainder)

X6. 301 (one more than a number in the 60 times table: 61, 121, 181, 241, 301; only 301 is in the 7 table: 7 × 43)

X7. 7, 14, 21, 28 and 35 (1 r 1, 2 r 2, 3 r 3, 4 r 4, 5 r 5; the remainder cannot be 6 or more)

X8. Wednesday (100 ÷ 7 = 14 r 2: after 14 weeks it is Monday again, then 2 more days)

X9. 7 pupils and 38 pens (giving 2 more pens each needs 10 + 4 = 14 pens, and 14 ÷ 2 = 7; 7 × 4 + 10 = 38, 7 × 6 = 42 = 38 + 4)

Lesson 6: Fraction Walls

C1. 5 (1/2 = 5/10)

C2. 1/6 (sixths are bigger pieces than ninths)

C3. 7/9

C4. 5/8 (8/8 – 3/8)

C5. 1/12, 5/12, 11/12

C6. 4 (2/5 = 4/10)

T1. 7/10 (3/5 = 6/10, and 7/10 is bigger than 6/10)

T2. She has read 3/4 (1/4 + 2/4). 1/4 is left.

T3. 7/12, 2/3, 5/6 (7/12, 8/12, 10/12)

T4. 2/3 and 8/12

T5. 1/3 (5/6 – 3/6 = 2/6 = 1/3)

T6. 3/8 (1/4 = 2/8 and 1/2 = 4/8)

T7. 7/8 (4/8 + 2/8 + 1/8)

T8. No. Both are missing one piece, but 2/3 is missing 1/3 and 3/4 is missing only 1/4, a smaller piece. So 3/4 is bigger (8/12 < 9/12).

X1. 5/9. 3/7 is less than 1/2 (half of 7 is 3 and a half) and 5/9 is more than 1/2 (half of 9 is 4 and a half).

X2. 4/9. 2/5 = 4/10, and ninths are bigger pieces than tenths, so 4/9 > 4/10.

X3. 1/2 = 3/6 and 1/3 = 2/6

X4. Mei. Ali 2/6, Ben 1/6, Mei 6/6 – 3/6 = 3/6 = 1/2.

X5. 9/12, 10/12 and 11/12 (2/3 = 8/12)

X6. 6/8 (3 + 4 = 7, and 14 is double 7, so double the top and the bottom)

X7. 8/12 (in 2/3 the bottom is 1 more than the top; 4 times as much: 8/12)

X8. 2 ways: 1/2 + 1/8 + 1/8 + 1/8 + 1/8, and 1/4 + 1/4 + 1/4 + 1/8 + 1/8.

X9. Siti, by 1/12 of a bottle (2/3 = 8/12, and 8/12 – 7/12 = 1/12).

Lesson 7: Area Architects

C1. 26 cm (9 + 4 + 9 + 4)

C2. 42 cm²

C3. 8 cm (32 ÷ 4)

C4. Each side is 6 cm (6 × 6 = 36), so the perimeter is 24 cm.

C5. Breadth 5 cm (8 × 5 = 40). Perimeter 26 cm.

C6. Area 7 cm². No: the perimeter depends on how the squares are joined (it can be 12, 14 or 16 cm).

T1. 24 by 1 (50 cm), 12 by 2 (28 cm), 8 by 3 (22 cm), 6 by 4 (20 cm). The 6 cm by 4 cm rectangle has the shortest perimeter.

T2. Length + breadth = 15, so the breadth is 3 cm. Area 12 × 3 = 36 cm².

T3. Perimeter 20 cm, so the square has sides of 5 cm. Square 25 cm², rectangle 16 cm². The square, by 9 cm².

T4. Each rectangle is 6 cm by 3 cm, so its perimeter is 18 cm.

T5. 5 tiles along and 4 tiles down: 20 tiles.

T6. 6 cm and 3 cm (6 × 3 = 18 and 6 + 3 + 6 + 3 = 18).

T7. Both fences are 28 m. Ali's garden has the bigger area: 49 m² against 45 m².

T8. 10 cm: only the 2 by 3 rectangle. 14 cm: any shape with no 2 by 2 block, e.g. a row of 6. 14 cm is the biggest: 6 squares have 24 sides, they need at least 5 touches to join, and each touch hides 2 sides (24 – 10 = 14).

X1. 11 by 1, 10 by 2, 9 by 3, 8 by 4, 7 by 5, 6 by 6: 6 rectangles. The 6 cm by 6 cm square has the biggest area, 36 cm².

X2. The perimeter is 6 sides of a small square, so each side is 6 cm. Area 2 × 36 = 72 cm².

X3. Yes, two answers. A cut parallel to the 4 cm sides is 4 cm long: 20 + 4 + 4 = 28 cm. A cut parallel to the 6 cm sides is 6 cm long: 20 + 6 + 6 = 32 cm.

X4. Small side 5 cm, big side 10 cm, perimeter 40 cm.

X5. For example: a 2 by 4 rectangle with one corner square missing (12 cm); a row of 7 squares (16 cm).

X6. Try breadths: 1 and 4 (area 4), 2 and 8 (16), 3 and 12 (36). So 12 cm by 3 cm, perimeter 30 cm.

X7. The perimeter is 8 sides of a small square, so each side is 32 ÷ 8 = 4 cm. Area 3 × 16 = 48 cm².

X8. Biggest 24 cm (12 sides of 2 cm, e.g. a row of 5). Smallest 20 cm (10 sides: a 2 by 2 square with one more square on a side).

X9. Area 35 – 4 = 31 cm². Perimeter still 24 cm: push the cut corners out and you get the 7 by 5 rectangle back.

Lesson 8: Angle and Line Detectives

C1. 4

C2. A right angle

C3. Smaller

C4. For example H and E (also F, N, Z, U, M).

C5. 4

C6. 3 (2 small ones and 1 made of both)

T1. A rectangle. 2 pairs of parallel lines.

T2. 6 (it has 8 corners; the 2 corners at the bottom of the gap are bigger than a right angle inside the shape).

T3. 3 o'clock and 9 o'clock

T4. 2 pairs of parallel lines and 4 pairs of perpendicular lines.

T5. (a) For example a shape with a long bottom, a shorter top parallel to it, one upright side and one slanted side. (b) For example a slanted parallelogram or a rhombus with no square corners.

T6. 4: 1 at the top corner, 2 where the middle line meets the upright line, 1 at the bottom corner.

T7. Yes, in both the hands are 4 hour-marks apart the short way round.

T8. 2 pairs of parallel lines and 4 pairs of perpendicular lines.

X1. 5 lines (4 + 3 + 2 + 1 = 10).

X2. 1 right angle; 3 right angles.

X3. 36: 4 corners × 1, 8 points on the sides × 2, and 4 crossing points × 4.

X4. 3: 1 at the top corner and 2 where the middle line meets the upright line.

X5. 6 pairs: 3 pairs of parallel lines (the 3 short lines) and 3 pairs of perpendicular lines (the upright line with each short line).

X6. No. A slanted parallelogram (or a rhombus) has 2 pairs of parallel lines but no right angles.

X7. The 4th angle is a right angle too (it is a rectangle). Yes: cut one corner off a rectangle with one slanted cut.

X8. 24. 4 at the corners, and each of the 5 lines makes 2 right angles at the top and 2 at the bottom: 4 + 5 × 4 = 24.

X9. No. The most is 5, for example an L shape. Children try to draw 6 square corners and always find a dent. A proof is not expected.

Lesson 9: Measure and Money Masters

C1. 245 cm

C2. 3 kg 60 g

C3. 2 ℓ 750 ml (4500 ml − 1750 ml = 2750 ml)

C4. $4.75 ($3.80 + $1.45 = $5.25; $10 − $5.25 = $4.75)

C5. 18:20

C6. 1 h 20 min (15 min + 1 h + 5 min)

T1. 6 cups, 200 ml left (300, 600, 900, 1200, 1500, 1800 ml; 2000 − 1800 = 200 ml)

T2. 11 h 30 min (30 min to 10:00, then 11 h to 21:00)

T3. 200 g (1050 g − 250 g = 800 g; 800 ÷ 4 = 200 g)

T4. The 07:55 bus (buses: 06:40, 07:05, 07:30, 07:55, 08:20; 07:55 + 35 min = 08:30; the 08:20 bus arrives at 08:55, too late).

T5. Four 50¢ coins and five 20¢ coins ($2 + $1 = $3)

T6. 2 h 35 min (20 min to 23:00, 1 h to 00:00, 1 h to 01:00, 15 min to 01:15)

T7. 5 km (1250 m × 4 = 5000 m)

T8. Two 1 ℓ cartons: $7.20 (one 1 ℓ and four small: $8.00; eight small: $8.80)

T9. 8 h 45 min (15 min to 22:00, 8 h to 06:00, 30 min to 06:30)

X1. 2 m 10 cm and 2 m 90 cm (500 − 80 = 420; 420 ÷ 2 = 210)

X2. Fill the 300 ml cup and pour it into the 500 ml cup. Fill the 300 ml cup again and pour until the 500 ml cup is full (200 ml goes in). 100 ml is left in the 300 ml cup.

X3. 10 erasers ($10 − $1.50 = $8.50 = 850¢; 10 × 85¢ = 850¢)

X4. 750 ml (3 × 900 = 2700 ml; 4200 − 2700 = 1500 ml; 1500 ÷ 2 = 750 ml)

X5. 11:30 (5 × 35 = 175 min, + 20 = 195 min = 3 h 15 min)

X6. $13.80 ($24 − $3.60 = $20.40; half is $10.20; Ali had $10.20 + $3.60 = $13.80, Ben $10.20)

X7. 300 g (half the water = 1200 − 750 = 450 g; all the water = 900 g; bottle = 1200 − 900 = 300 g)

X8. $1.60 (take away the extra 2 × 60¢ = $1.20: $8.00 for 5 pens; 800¢ ÷ 5 = 160¢. A file is $2.20.)

X9. 14:35, which is 2:35 p.m. (real start 12:55; + 5 min = 13:00, + 1 h = 14:00, + 35 min = 14:35)

Lesson 10: Challenge Day

C1. 8 (5 + 4 − 1)

C2. 19 and 31 (50 − 12 = 38; 38 ÷ 2 = 19; 19 + 12 = 31)

C3. 147, 174, 417, 471, 714, 741. Even: 174 and 714.

C4. 35

C5. 41 (add 4 each time: 5, 9, 13, 17, 21, 25, 29, 33, 37, 41)

C6. $2.30 ($1.35 + $1.35 = $2.70; $5 − $2.70 = $2.30)

T1. 12 (4 + 1 + 7)

T2. Ali: fish. Ben: cat. Siti: dog.

T3. 25 sticks (add 2 each time: 3 + 11 × 2 = 25)

T4. Largest even: 9630. Smallest odd: 3069.

T5. 145 (Mei 35, Siti 55; 35 + 20 + 35 + 55 = 145)

T6. Siti (the order is Siti, Ben, Ali)

T7. Day 4 (highest point each day: 5, 8, 11, 14 m; it passes 12 m on day 4)

T8. 62 and 38

X1. Mei 25, Ali 35, Ravi 30

X2. 14 (9 small, 4 of size 2 by 2, 1 of size 3 by 3)

X3. 15 (5 + 4 + 3 + 2 + 1)

X4. 9 (18, 27, 36, 45, 54, 63, 72, 81, 90)

X5. Ali 10, Ben 10, Mei 13, Siti 11

X6. Three 50¢ coins and five 10¢ coins

X7. Mei (if Ali ate it, Ben and Mei are both truthful; if Ben ate it, Ali and Mei are both truthful; if Mei ate it, only Ben is truthful)

X8. 13 (Siti 6th, Ravi 6 + 3 + 1 = 10th from the front, then 3 more behind Ravi)

X9. 3P1 40 kg, 3P2 30 kg, 3P3 40 kg, 3P4 25 kg (take away the extras 10 + 10 − 5 = 15: 120 kg ÷ 4 = 30 kg for 3P2)